AIS – AURELIA INDUSTRIAL SYSTEMS

PREVENTIVE MAINTENANCE GUIDE

AIS Preventive Maintenance Guide
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1. Document Purpose

This document defines the preventive maintenance requirements for AIS industrial assembly systems.

It provides general maintenance intervals, recommended activities, safety requirements, and guidelines for determining whether preventive maintenance is due.

This document applies to the AIS-100, AIS-200, and AIS-500 product families.

Machine-specific maintenance procedures may contain additional requirements and must be followed where applicable.

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2. Preventive Maintenance Philosophy

Preventive maintenance is performed to reduce the probability of unexpected machine failures and maintain safe and reliable operation.

Maintenance intervals are based primarily on accumulated machine operating hours.

Customers should also consider:

- machine operating conditions;
- production intensity;

- environmental conditions;
- previous failures;
- maintenance history;
- machine-specific recommendations.

The operating-hour interval defined in the **current approved documentation** must be used when determining the standard preventive maintenance schedule.

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3. Current Maintenance Intervals

The current standard preventive maintenance intervals are:

Product	Preventive Maintenance Interval
AIS-100	Every 1,500 operating hours
AIS-200	Every 2,000 operating hours
AIS-500	Every 2,500 operating hours

These intervals represent the current standard values as of January 2026.

Previous documentation may contain different intervals. When conflicting versions are found, the current approved maintenance documentation takes precedence.

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4. AIS-100 Maintenance

The standard AIS-100 preventive maintenance interval is:

Every 1,500 operating hours

Recommended activities include:

- mechanical inspection;
- inspection of moving components;
- sensor inspection;
- inspection of electrical connections;
- safety-device verification;

- cleaning;
- actuator inspection;
- functional testing.

The maintenance technician should record the current machine operating hours and all activities performed.

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5. AIS-200 Maintenance

The standard AIS-200 preventive maintenance interval is:

Every 2,000 operating hours

Recommended activities include:

- mechanical inspection;
- robotic-system inspection;
- sensor inspection;
- electrical inspection;
- safety-system verification;
- cooling-system inspection;
- cleaning;
- actuator inspection;
- functional testing;
- diagnostic review.

5.1 Cooling-System Inspection

For AIS-200 systems, preventive maintenance should include inspection of the cooling system.

The technician should verify:

- cooling-system operating status;
- coolant condition where applicable;
- coolant level where applicable;

- visible leakage;
- abnormal temperatures;
- cooling-unit alarms;
- relevant diagnostic information.

A recurring cooling-system alarm should be investigated rather than simply cleared.

5.2 Error E-204 and Maintenance

Error **E-204** indicates that the cooling-system temperature has exceeded the configured threshold.

If E-204 occurs repeatedly, the condition should be investigated as part of technical troubleshooting and, where appropriate, maintenance activity.

Operators must not bypass the temperature protection or modify the configured threshold.

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6. AIS-500 Maintenance

The standard AIS-500 preventive maintenance interval is:

Every 2,500 operating hours

Recommended activities include:

- robotic-system inspection;
- mechanical inspection;
- sensor inspection;
- safety-system verification;
- electrical inspection;
- actuator inspection;
- communication-system verification;
- cleaning;
- functional testing;
- diagnostic review.

Due to the higher production throughput and multiple robotic modules, maintenance planning should consider production load and machine operating conditions.

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7. Determining Whether Maintenance Is Due

Maintenance status should be determined by comparing:

1. the machine's current operating hours;
2. the applicable product maintenance interval;
3. the machine's maintenance history.

Example

An AIS-200 has:

- current operating hours: 1,734 hours;
- current maintenance interval: 2,000 hours;
- most recent preventive maintenance: 1,500 hours.

The machine has not yet reached the next standard maintenance threshold.

The remaining operating hours before the next threshold are:

2,000 − 1,734 = 266 operating hours

The result should be interpreted together with machine condition and customer maintenance procedures.

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8. Maintenance History

Maintenance activities should be recorded with at least:

- machine serial number;
- product model;
- maintenance type;
- operating hours at intervention;
- intervention date;
- activities performed;

- replaced components;
- findings;
- technician information;
- follow-up actions.

Maintenance records must be associated with the correct customer and machine.

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9. Overdue Maintenance

A machine is considered overdue when its current operating hours exceed the applicable maintenance threshold and the corresponding preventive maintenance has not been completed.

An overdue machine should be reviewed by qualified maintenance personnel.

Example

An AIS-500 has:

- current operating hours: 5,020;
- standard maintenance interval: 2,500 hours.

If the required maintenance has not been performed according to the applicable maintenance history, the machine requires maintenance review.

Operating hours alone are not sufficient to determine the complete maintenance history.

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10. Safety Requirements

Before performing maintenance:

- stop the machine according to the applicable shutdown procedure;
- isolate energy sources where required;
- follow applicable lockout/tagout procedures;
- verify that hazardous motion has stopped;
- use appropriate personal protective equipment;
- restrict access to authorized personnel.

Safety devices must not be bypassed during maintenance unless an approved procedure explicitly requires a controlled diagnostic operation.

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11. Maintenance Escalation

Technical Support should be contacted when:

- a maintenance procedure cannot be completed;
- a safety-related problem is identified;
- a recurring alarm remains unresolved;
- unexpected component wear is found;
- the machine cannot safely return to operation;
- diagnostic information indicates a potential technical failure.

Engineering escalation may be required for repeated or complex failures.

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12. AI Assistant Usage

The AIS AI Support Assistant may:

- retrieve the applicable maintenance interval;
- retrieve maintenance records through authorized tools;
- calculate approximate remaining operating hours;
- explain documented maintenance procedures;
- identify potentially overdue maintenance based on authorized data.

The AI assistant must not:

- authorize maintenance work;
- declare a machine safe to operate;
- override safety procedures;
- modify maintenance records without authorization;
- approve or reject warranty coverage;
- replace qualified technical personnel.

13. Document Control

Version	Date	Description
1.0	2025-01-15	Initial maintenance guide
1.5	2025-07-10	Updated product maintenance information
2.0	2026-01-20	Current maintenance intervals and procedures

Current approved version: 2.0